

Nivetha Lakshmi | Sr. Data Engineer

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PROFESSIONAL SUMMARY:

Dedicated **IT professional** with **8+ years** of experience in developing/providing multiple data related solutions, specializing in integrating source data using **Azure Data Factory** and processing it with **Databricks**. Proficient in creating, maintaining, and ADF mappings/pipelines. I have a deep understanding of **ETL concepts** and data modeling. Expert in writing and **optimizing complex SQL queries**, and good in **SQL performance tuning**. Have strong **problem-solving skills**, I excel at **troubleshooting and debugging issues** to ensure data integration and for advanced data analytics in Agile project delivery mode.

- Specialized in ETL processes, data integration, and cloud-based data platforms like **Azure** and **Databricks**.
- Expert in designing, implementing, and optimizing **Azure Data Factory (ADF)** mappings for data solutions.
- Extensive experience in **Databricks** for large dataset processing using **PySpark** and **Scala**.
- Strong database management skills with **SQL Server**, **Oracle**, **MySQL**, and **Azure SQL**.
- Proficient in **Dimensional Modeling**, EDW concepts, and implementing **Change Data Capture (CDC)**.
- Skilled in gathering **Business requirements**, finalizing **source-to-target mappings**, & preparing technical docs.
- Strong expertise in integrating data from heterogeneous sources like **databases**, **flat files**, **JSON**, and **APIs** into scalable storage solutions using **Azure Data Lake Storage (ADLS Gen2)**.
- Experienced in developing ETL solutions using ADF components such as **Linked Services**, **Key Vault**, **Datasets**, **Mapping Data Flow**, **Triggers**, and **Alerts**.
- Solid understanding of using **Azure DevOps** for **CI/CD pipelines**, build, and release management.
- Extensive experience with **PowerCenter** for efficient data integration.
- Expertise in **SQL optimization**, creating complex queries, working with **ER diagrams**, **subqueries**, and **analytical functions**.
- Proficient in optimizing resource utilization in **Azure Data Factory** using **parallelism**, **data partitioning**, and minimizing redundancy in workflows.
- Collaborative team player with strong experience in **Agile environments**, ensuring timely project delivery.

TECHNICAL SKILLS

Core Competencies

- **ETL Tools:** Azure Data Factory (ADF), Databricks, Informatica PowerCenter
- **Data Storage & Databases:** Azure SQL, SQL Server, Snowflake, MySQL, Oracle 12c/19c
- **Cloud Platforms:** Microsoft Azure, ADLS Gen2, AWS S3
- **Programming & Frameworks:** PySpark, Scala, SQL, T-SQL
- **Data Integration & Modeling:** Delta Lake, Data Warehousing, Dimensional Modeling, CDC
- **DevOps & Tools:** Azure DevOps, GitHub, Control-M, JIRA, Confluence, SharePoint
- **Performance Optimization:** Query Tuning, Partitioning, Pushdown Optimization, Parallelism

PROFESSIONAL EXPERIENCE

Cigna, Austin, TX**Apr 2023 to Present****Sr. Data Engineer**

Cigna Corporation provides insurance and related products and services in the United States. This project name is **CDB (Commercials database)** which falls under Cigna Supplement Business, supplemental insurance policies help pay for many expenses that are not covered by primary health insurance, these policies can help to pay for out-of-pocket medical expenses such as copays and deductibles when a serious illness or accident happens. Cigna sources data from multiple sources using ADF tool into **Azure Data Lake** tables.

- Actively engaged in interacting with business users to capture requirements and conduct Business Analysis.
- Designed and implemented end-to-end ETL/ELT pipelines in Azure Data Factory to orchestrate data ingestion and transformation from diverse sources
- Integrated ADF with Azure Databricks, ADLS, SQL Databases, and other Azure services for seamless data processing.
- Configured event-based, tumbling window, and scheduled triggers to automate pipeline executions.
- Enabled CI/CD for ADF pipelines using Azure DevOps and ARM templates.
- Implemented error handling, retry logic, and custom logging using Azure Monitor/Log Analytics for proactive monitoring.
- Optimized pipeline performance by leveraging parallel processing, staging layers, and partitioned data loading.
- Developed data transformation and processing workflows in Azure Databricks using PySpark and Spark SQL.
- Implemented Delta Lake for efficient incremental loads, CDC (Change Data Capture), and time travel for historical data analysis.
- Optimized Spark jobs by tuning cluster configurations, caching strategies, and code optimization to improve runtime performance
- Integrated Databricks with ADF as an orchestration layer and ADLS as a data lake for storage and staging.
- Implemented data quality checks and validation frameworks within Databricks notebooks.
- Proficient in Spark optimization techniques, broadcast joins, partitioning, and caching to enhance job performance.
- Participated in daily scrum, sprint planning, grooming, and retrospective meetings as part of the Agile process.

Environment: ADF, Databricks, PySpark, SQL, Oracle 18c, SQL Server 2017, Oracle SQL Developer, TOAD, Control M, Bitbucket

Google, Austin, TX**Aug 2022 to Mar 2023****Data Analyst**

The objective of the project is to leverage the onboarding experience of Nooglers who are onboarded for both Sales and Technical teams. Cloud Sales onboarding is part of the larger Sales Excellence and Productivity (SE&P) organization, which provides impactful strategies based on insights delivered through programs, services, best practices, and methodologies to enable the field to win more business for Google Cloud.

- Analyzed the Nooglers Cloud Onboarding data to derive meaningful insights.
- Perform drill down analysis on the data to retrieve top contributors for Facilitators & Volunteering categories.
- Analyzed **business** and **system requirements** to identify **KPI's and metrics**.
- Applied generic formulas and embedded them in the dashboard to make it reusable across Fiscal quarters.
- Interacted with the Cloud Onboarding team on data discrepancy to resolve it.
- Provided necessary updates to project team to have necessary Action & flagged any risks upfront.

- Provided guidance and data insights to new team members.
- Populated KPI metrics for it to be utilized in Tableau.
- Assisted in data visualization and customization for Nooglers in Tableau.

Environment: Google Cloud Platform (GCP), SQL Server, ORACLE 18c, Tableau.

Curative, Austin, TX

Sept 2020 to Jul 2022

Data Analyst/ETL Dev.

Curative is a health insurance company founded in 2020 that provides employer-based health plans with a unique \$0 deductible, \$0 copay model for in-network care. Headquartered in Austin, Texas, the company is a relatively new player that gained prominence. It offers a flat, monthly premium with no additional out-of-pocket costs for in-network care, including copays and deductibles.

- Designed and implemented end-to-end data pipelines in Azure Data Factory to orchestrate data ingestion, transformation, and loading (ETL/ELT) across multiple data sources
- Configured and optimized triggers (schedule, tumbling window, event-based) for automated pipeline executions.
- Implemented copy activities and mapping data flows to integrate data from on-premises and cloud sources (SQL Server, Blob Storage, ADLS, REST APIs, etc.).
- Handled error handling and fault tolerance with custom logging, retry policies, and alerts via Azure Monitor/Log Analytics.
- Collaborated with DevOps teams to enable CI/CD for ADF pipelines using ARM templates and Azure DevOps.
- Improved pipeline performance by optimizing data partitioning, parallelism, and leveraging staging techniques in data lake.
- Designed and developed dimension and fact tables in Azure Synapse Dedicated SQL Pool to support enterprise data warehouse and reporting requirements.
- Performed data loading into Synapse using PolyBase, COPY command, and ADF integration for high-throughput ingestion
- Applied best practices for table distribution (hash, replicate) and indexing (clustered columnstore) to optimize storage and query execution.
- Developed ETL/ELT pipelines to populate dimensions (slowly changing dimensions – SCD Type 1 & Type 2) and fact tables using Azure Data Factory.
- Tuned queries and improved performance by analyzing query execution plans, statistics
- Knowledge of monitoring tools like Azure Monitor, Log Analytics, and cost optimization strategies in Synapse.
- Exposure to Azure Key Vault, Managed Identities, and secure data handling.
- Integrated Synapse with Power BI, ADF, and Azure Data Lake Storage (ADLS) for seamless analytics workflows.

Environment: Informatica PowerCenter 9.6.1, SQL Server 2019, ORACLE 18c, SQL, UNIX, Toad, Snowflake Cloud DB, SQL Developer, AutoSys, JIRA

Union Bank, Dallas, TX

Jan 2018 to Aug 2020

Informatica Developer

Description: Union Bank that provides retail, commercial, and municipal banking products, and services. As a part of this project, data is being collected from all servicing originations Application systems, databases like oracle, flat files, third party. These data are being integrated into **Integrated Ware House** after applying business rules thru ETL tools Informatica.

- Analyze business requirements, technical specification, source repositories and physical data models for ETL mapping and process flow.
- Worked extensively with mappings using expressions, aggregators, filters, lookup, joiners, update strategy and stored procedure transformations.
- Developed mapping to load Fact and Dimension tables, for type 1 and type 2 dimensions and incremental loading and unit tested the mappings.
- Coordinate and developed all documents related to ETL design and development.
- Responsible for debugging and performance tuning of targets, sources, mappings and sessions.
- Optimized the mappings and implementing the complex business rules by creating re-usable transformations and Mapplets.
- Optimized the source queries in order to control the temp space and added delay intervals depending upon the business requirement for performance.
- Used Informatica workflow manager for creating, running the Batches and Sessions and scheduling them to run at specified time.
- Executed sessions, sequential and concurrent batches for proper execution of mappings and set up email delivery after execution.
- Implemented and documented all the best practices used for the data warehouse.
- Improving the performance tuning of ETL mapping/workflow
- Created Workflows, tasks, database connections using workflow manager.
- Responsible for identifying bugs in existing mappings by analyzing data flow, evaluating transformations and fixing bugs.
- Code walks through with team members.
- Involved in testing the database using complex SQL scripts and handled the performance issues effectively.

Environment: Informatica Power Center 9.5, Oracle 12c, SQL, SQL Server 2017, JIRA

EDUCATION:

- **B. Tech in IT from Anna University, Chennai, INDIA**

CERTIFICATIONS:

- **Microsoft Certified: Azure Data Fundamentals (Exam DP-900)**
- **Microsoft Certified: Fabric Data Engineer Associate (Exam DP-700)**